Neural Computation -- 1994

79 article(s), all issues

1. Cortical Map Reorganization as a Competitive Process

Granger Sutton, James A. Reggia, Steven Armentrout, C. Lynne D'Autrechy

https://doi.org/10.1162/neco.1994.6.1.1 | pp. 1-13

Past models of somatosensory cortex have successfully demonstrated map formation and subsequent map reorganization following localized repetitive stimuli or deafferentation. They provide an impressive demonstration that fairly simple assumptions about cortical connectivity and synaptic plasticity can account for several observations concerning cortical maps. However, past models have not successfully demonstrated spontaneous map reorganization following cortical lesions. Recently, an assumption universally used in these and other cortex models, that peristimulus inhibition is due solely to horizontal intracortical inhibitory connections, has been questioned and an additional mechanism, the competitive distribution of activity, has been proposed. We implemented a computational model of somatosensory cortex based on competitive distribution of activity. This model exhibits spontaneous map reorganization in response to a cortical lesion, going through a two-phase reorganization process. These results make a testable prediction that can be used to experimentally support or refute part of the competitive distribution hypothesis, and may lead to practically useful computational models of recovery following stroke.

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2. An Efficient Method for Computing Synaptic Conductances Based on a Kinetic Model of Receptor Binding

Alain Destexhe, Zachary F. Mainen, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1994.6.1.14 | pp. 14-18

International audience

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3. A Neural Network for Coding of Trajectories by Time Series of Neuronal Population Vectors

A. V. Lukashin, Apostolos P. Georgopoulos

https://doi.org/10.1162/neco.1994.6.1.19 | pp. 19-28

The neuronal population vector is a measure of the combined directional tendency of the ensemble of directionally tuned cells in the motor cortex. It has been found experimentally that a trajectory of limb movement can be predicted by adding together population vectors, tip-to-tail, calculated for successive instants of time to construct a neural trajectory. In the present paper we consider a model of the dynamic evolution of the population vector. The simulated annealing algorithm was used to adjust the connection strengths of a feedback neural network so that it would generate a given trajectory by a sequence of population vectors. This was repeated for different trajectories. Resulting sets of connection strengths reveal a common feature regardless of the type of trajectories generated by the network: namely, the mean connection strength was negatively correlated with the angle between the preferred directions of neuronal pair involved in the connection. The results are discussed in the light of recent experimental findings

concerning neuronal connectivity within the motor cortex.

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4. Theoretical Considerations for the Analysis of Population Coding in Motor Cortex

Terence D. Sanger

https://doi.org/10.1162/neco.1994.6.1.29 | pp. 29-37

Recent evidence of population coding in motor cortex has led some researchers to claim that certain variables such as hand direction or force may be coded within a Cartesian coordinate system with respect to extra personal space. These claims are based on the ability to predict the rectangular coordinates of hand movement direction using a “population vector” computed from multiple cells' firing rates. I show here that such a population vector can always be found given a very general set of assumptions. Therefore the existence of a population vector constitutes only weak support for the explicit use of a particular coordinate representation by motor cortex.

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5. Neural Network Model of the Cerebellum: Temporal Discrimination and the Timing of Motor Responses

Dean V. Buonomano, Michael D. Mauk

https://doi.org/10.1162/neco.1994.6.1.38 | pp. 38-55

Substantial evidence has established that the cerebellum plays an important role in the generation of movements. An important aspect of motor output is its timing in relation to external stimuli or to other components of a movement. Previous studies suggest that the cerebellum plays a role in the timing of movements. Here we describe a neural network model based on the synaptic organization of the cerebellum that can generate timed responses in the range of tens of milliseconds to seconds. In contrast to previous models, temporal coding emerges from the dynamics of the cerebellar circuitry and depends neither on conduction delays, arrays of elements with different time constants, nor populations of elements oscillating at different frequencies. Instead, time is extracted from the instantaneous granule cell population vector. The subset of active granule cells is time-varying due to the granule—Golgi—granule cell negative feedback. We demonstrate that the population vector of simulated granule cell activity exhibits dynamic, nonperiodic trajectories in response to a periodic input. With time encoded in this manner, the output of the network at a particular interval following the onset of a stimulus can be altered selectively by changing the strength of granule $\rightarrow$ Purkinje cell connections for those granule cells that are active during the target time window. The memory of the reinforcement at that interval is subsequently expressed as a change in Purkinje cell activity that is appropriately timed with respect to stimulus onset. Thus, the present model demonstrates that a network based on cerebellar circuitry can learn appropriately timed responses by encoding time as the population vector of granule cell activity.

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6. Computational Aspects of the Respiratory Pattern Generator

Allan Gottschalk, Malcolm D. Ogilvie, Diethelm W. Richter, Allan I Pack

https://doi.org/10.1162/neco.1994.6.1.56 | pp. 56-68

To help evaluate the hypothesis that the central respiratory rhythm is generated by a network of interacting neurons, a network model of respiratory rhythmogenesis is formulated and examined computationally. The neural elements of the network are driven by tonic inputs and generate a continuous variable representing firing rate. Each neural element in the model can be described by an activation time constant, an adaptation time constant, and a step nonlinearity. Initial network connectivity was based on an earlier proposed model of the central respiratory pattern generator. These connections were adjusted interactively until the model trajectories resembled those observed electrophysiologically. The properties of the resulting network were examined computationally by simulation, determination of the phase resetting behavior of the network oscillator, and examination of the localized eigenstructure of the network. These results demonstrate that the network model can account for a number of diverse physiological observations, and, thus, support the network hypothesis of respiratory rhymogenesis.

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7. Subharmonic Coordination in Networks of Neurons with Slow Conductances

Thomas LoFaro, Nancy Kopell, Eve Marder, Scott L. Hooper

https://doi.org/10.1162/neco.1994.6.1.69 | pp. 69-84

We study the properties of a network consisting of two model neurons that are coupled by reciprocal inhibition. The study was motivated by data from a pair of cells in the crustacean stomatogastric ganglion. One of the model neurons is an endogenous burster; the other is excitable but not bursting in the absence of phasic input. We show that the presence of a hyperpolarization activated inward current (i h) in the excitable neuron allows these neurons to fire in integer subharmonics, with the excitable cell firing once for every N ≥ 1 bursts of the oscillator. The value of N depends on the amount of hyperpolarizing current injected into the excitable cell as well as the voltage activation curve of i h . For a fast synapse, these parameter changes do not affect the characteristic point in the oscillator cycle at which the excitable cell bursts; for slower synapses, such a relationship is maintained within small windows for each N. The network behavior in the current work contrasts with the activity of a pair of coupled oscillators for which the interaction is through phase differences; in the latter case, subharmonics exist if the uncoupled oscillators have near integral frequency relationships, but the phase relationships of the oscillators in general change significantly with parameters. The mechanism of this paper provides a potential means of coordinating subnetworks acting on different time scales but maintaining fixed relationships between characteristic points of the cycles.

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8. Setting the Activity Level in Sparse Random Networks

Ali A. Minai, William B. Levy

https://doi.org/10.1162/neco.1994.6.1.85 | pp. 85-99

We investigate the dynamics of a class of recurrent random networks with sparse, asymmetric excitatory connectivity and global shunting inhibition mediated by a single interneuron. Using probabilistic arguments and a hyperbolic tangent approximation to the gaussian, we develop a simple method for setting the average level of firing activity in these networks. We demonstrate through simulations that our technique works well and extends to networks with more complicated inhibitory schemes. We are interested primarily in the CA3 region of the mammalian hippocampus, and the random networks investigated here are seen as modeling the a priori dynamics of activity in this region. In the presence of external stimuli, a suitable

synaptic modification rule could shape this dynamics to perform temporal information processing tasks such as sequence completion and prediction.

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9. The Role of Constraints in Hebbian Learning

Kenneth D. Miller, David Mackay

https://doi.org/10.1162/neco.1994.6.1.100 | pp. 100-126

Models of unsupervised, correlation-based (Hebbian) synaptic plasticity are typically unstable: either all synapses grow until each reaches the maximum allowed strength, or all synapses decay to zero strength. A common method of avoiding these outcomes is to use a constraint that conserves or limits the total synaptic strength over a cell. We study the dynamic effects of such constraints. Two methods of enforcing a constraint are distinguished, multiplicative and subtractive. For otherwise linear learning rules, multiplicative enforcement of a constraint results in dynamics that converge to the principal eigenvector of the operator determining unconstrained synaptic development. Subtractive enforcement, in contrast, typically leads to a final state in which almost all synaptic strengths reach either the maximum or minimum allowed value. This final state is often dominated by weight configurations other than the principal eigenvector of the unconstrained operator. Multiplicative enforcement yields a “graded” receptive field in which most mutually correlated inputs are represented, whereas subtractive enforcement yields a receptive field that is “sharpened” to a subset of maximally correlated inputs. If two equivalent input populations (e.g., two eyes) innervate a common target, multiplicative enforcement prevents their segregation (ocular dominance segregation) when the two populations are weakly correlated; whereas subtractive enforcement allows segregation under these circumstances. These results may be used to understand constraints both over output cells and over input cells. A variety of rules that can implement constrained dynamics are discussed.

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10. Toward a Theory of the Striate Cortex

Li Zhaoping, Joseph J. Atick

https://doi.org/10.1162/neco.1994.6.1.127 | pp. 127-146

We explore the hypothesis that linear cortical neurons are concerned with building a particular type of representation of the visual world—one that not only preserves the information and the efficiency achieved by the retina, but in addition preserves spatial relationships in the input—both in the plane of vision and in the depth dimension. Focusing on the linear cortical cells, we classify all transforms having these properties. They are given by representations of the scaling and translation group and turn out to be labeled by rational numbers ‘(p + q)/p’ (p, q integers). Any given (p, q) predicts a set of receptive fields that comes at different spatial locations and scales (sizes) with a bandwidth of log 2 [(p + q)/p] octaves and, most interestingly, with a diversity of ‘q’ cell varieties. The bandwidth affects the trade-off between preservation of planar and depth relations and, we think, should be selected to match structures in natural scenes. For bandwidths between 1 and 2 octaves, which are the ones we feel provide the best matching, we find for each scale a minimum of two distinct cell types that reside next to each other and in phase quadrature, that is, differ by 90° in the phases of their receptive fields, as are found in the cortex, they resemble the “even-symmetric” and “odd-symmetric” simple cells in special cases. An interesting consequence of the representations presented here is that the pattern of activation in the cells in response to a translation or scaling of an object remains the same but merely shifts its locus from one group of cells to another. This work also provides a new

understanding of color coding changes from the retina to the cortex.

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11. Fast Exact Multiplication by the Hessian

Barak A. Pearlmutter

https://doi.org/10.1162/neco.1994.6.1.147 | pp. 147-160

Just storing the Hessian H (the matrix of second derivatives δ 2 E/δw i δw j of the error E with respect to each pair of weights) of a large neural network is difficult. Since a common use of a large matrix like H is to compute its product with various vectors, we derive a technique that directly calculates Hv, where v is an arbitrary vector. To calculate Hv, we first define a differential operator R v {f(w)} = (δ/δr)f(w + rv)|r=0, note that R v {■w} = Hv and R v {w} = v, and then apply R v {·} to the equations used to compute ■ w . The result is an exact and numerically stable procedure for computing Hv, which takes about as much computation, and is about as local, as a gradient evaluation. We then apply the technique to a one pass gradient calculation algorithm (backpropagation), a relaxation gradient calculation algorithm (recurrent backpropagation), and two stochastic gradient calculation algorithms (Boltzmann machines and weight perturbation). Finally, we show that this technique can be used at the heart of many iterative techniques for computing various properties of H, obviating any need to calculate the full Hessian.

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12. Polyhedral Combinatorics and Neural Networks

Andrew H. Gee, Richard W. Prager

https://doi.org/10.1162/neco.1994.6.1.161 | pp. 161-180

The often disappointing performance of optimizing neural networks can be partly attributed to the rather ad hoc manner in which problems are mapped onto them for solution. In this paper a rigorous mapping is described for quadratic 0-1 programming problems with linear equality and inequality constraints, this being the most general class of problem such networks can solve. The problem's constraints define a polyhedron P containing all the valid solution points, and the mapping guarantees strict confinement of the network's state vector to P. However, forcing convergence to a 0-1 point within P is shown to be generally intractable, rendering the Hopfield and similar models inapplicable to the vast majority of problems. A modification of the tabu learning technique is presented as a more coherent approach to general problem solving with neural networks. When tested on a collection of knapsack problems, the modified dynamics produced some very encouraging results.

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13. Hierarchical Mixtures of Experts and the EM Algorithm

Michael I. Jordan, Robert A. Jacobs

https://doi.org/10.1162/neco.1994.6.2.181 | pp. 181-214

We present a tree-structured architecture for supervised learning. The statistical model underlying the architecture is a hierarchical mixture model in which both the mixture coefficients and the mixture components are generalized linear models (GLIM's). Learning is treated as a maximum likelihood problem; in particular, we present an Expectation-Maximization (EM) algorithm for adjusting the parameters of the architecture. We also develop an on-line learning algorithm in which the parameters are updated

incrementally. Comparative simulation results are presented in the robot dynamics domain.

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14. TD-Gammon, a Self-Teaching Backgammon Program, Achieves Master-Level Play

Gerald Tesauro

https://doi.org/10.1162/neco.1994.6.2.215 | pp. 215-219

TD-Gammon is a neural network that is able to teach itself to play backgammon solely by playing against itself and learning from the results, based on the TD(λ) reinforcement learning algorithm (Sutton 1988). Despite starting from random initial weights (and hence random initial strategy), TD-Gammon achieves a surprisingly strong level of play. With zero knowledge built in at the start of learning (i.e., given only a “raw” description of the board state), the network learns to play at a strong intermediate level. Furthermore, when a set of hand-crafted features is added to the network's input representation, the result is a truly staggering level of performance: the latest version of TD-Gammon is now estimated to play at a strong master level that is extremely close to the world's best human players.

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15. Correlated Attractors from Uncorrelated Stimuli

Leticia F. Cugliandolo

https://doi.org/10.1162/neco.1994.6.2.220 | pp. 220-224

Analytic results for a neural network model proposed to account for neurophysiological experiments in which attractors for uncorrelated patterns are found to be correlated are presented. Exact expressions for the associative attractors and their correlations are described.

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16. Learning of Phase Lags in Coupled Neural Oscillators

Bard Ermentrout, Nancy Kopell

https://doi.org/10.1162/neco.1994.6.2.225 | pp. 225-241

If an oscillating neural circuit is forced by another such circuit via a composite signal, the phase lag induced by the forcing can be changed by changing the relative strengths of components of the coupling. We consider such circuits, with the forced and forcing oscillators receiving signals with some given phase lag. We show how such signals can be transformed into an algorithm that yields connection strengths needed to produce that lag. The algorithm reduces the problem of producing a given phase lag to one of producing a kind of synchrony with a “teaching” signal; the algorithm can be interpreted as maximizing the correlation between voltages of a cell and the teaching signal. We apply these ideas to regulation of phase lags in chains of oscillators associated with undulatory locomotion.

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17. A Mechanism for Neuronal Gain Control by Descending Pathways

Mark Nelson

https://doi.org/10.1162/neco.1994.6.2.242 | pp. 242-254

Many implementations of adaptive signal processing in the nervous system are likely to require a mechanism for gain control at the single neuron level. To properly adjust the gain of an individual neuron, it may be necessary to use information carried by neurons in other parts of the system. The ability to adjust the gain of neurons in one part of the brain, using control signals arising from another, has been observed in the electrosensory system of weakly electric fish, where descending pathways to a first-order sensory nucleus have been shown to influence the gain of its output neurons. Although the neural circuitry associated with this system is well studied, the exact nature of the gain control mechanism is not fully understood. In this paper, we propose a mechanism based on the regulation of total membrane conductance via synaptic activity on descending pathways. Using a simple neural model, we show how the activity levels of paired excitatory and inhibitory control pathways can regulate the gain and baseline excitation of a target neuron.

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18. The Role of Weight Normalization in Competitive Learning

Geoffrey J. Goodhill, Harry G. Barrow

https://doi.org/10.1162/neco.1994.6.2.255 | pp. 255-269

The effect of different kinds of weight normalization on the outcome of a simple competitive learning rule is analyzed. It is shown that there are important differences in the representation formed depending on whether the constraint is enforced by dividing each weight by the same amount (“divisive enforcement”) or subtracting a fixed amount from each weight (“subtractive enforcement”). For the divisive cases weight vectors spread out over the space so as to evenly represent “typical” inputs, whereas for the subtractive cases the weight vectors tend to the axes of the space, so as to represent “extreme” inputs. The consequences of these differences are examined.

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19. A Probabilistic Resource Allocating Network for Novelty Detection

Stephen Roberts, Lionel Tarassenko

https://doi.org/10.1162/neco.1994.6.2.270 | pp. 270-284

The detection of novel or abnormal input vectors is of importance in many monitoring tasks, such as fault detection in complex systems and detection of abnormal patterns in medical diagnostics. We have developed a robust method for novelty detection, which aims to minimize the number of heuristically chosen thresholds in the novelty decision process. We achieve this by growing a gaussian mixture model to form a representation of a training set of “normal” system states. When previously unseen data are to be screened for novelty we use the same threshold as was used during training to define a novelty decision boundary. We show on a sample problem of medical signal processing that this method is capable of providing robust novelty decision boundaries and apply the technique to the detection of epileptic seizures within a data record.

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20. Diffusion Approximations for the Constant Learning Rate Backpropagation Algorithm and Resistance to Local Minima

William Finnoff

https://doi.org/10.1162/neco.1994.6.2.285 | pp. 285-295

In this paper we discuss the asymptotic properties of the most commonly used variant of the backpropagation algorithm in which network weights are trained by means of a local gradient descent on examples drawn randomly from a fixed training set, and the learning rate η of the gradient updates is held constant (simple backpropagation). Using stochastic approximation results, we show that for $\eta \to 0$ this training process approaches a batch training. Further, we show that for small η one can approximate simple backpropagation by the sum of a batch training process and a gaussian diffusion, which is the unique solution to a linear stochastic differential equation. Using this approximation we indicate the reasons why simple backpropagation is less likely to get stuck in local minima than the batch training process and demonstrate this empirically on a number of examples.

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21. Relating Real-Time Backpropagation and Backpropagation-Through-Time: An Application of Flow Graph Interreciprocity

Françoise Beaufays, Eric A. Wan

https://doi.org/10.1162/neco.1994.6.2.296 | pp. 296-306

We show that signal flow graph theory provides a simple way to relate two popular algorithms used for adapting dynamic neural networks, real-time backpropagation and backpropagation-through-time. Starting with the flow graph for real-time backpropagation, we use a simple transposition to produce a second graph. The new graph is shown to be interreciprocal with the original and to correspond to the backpropagation-through-time algorithm. Interreciprocity provides a theoretical argument to verify that both flow graphs implement the same overall weight update.

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22. Smooth On-Line Learning Algorithms for Hidden Markov Models

Pierre Baldi, Yves Chauvin

https://doi.org/10.1162/neco.1994.6.2.307 | pp. 307-318

A simple learning algorithm for Hidden Markov Models (HMMs) is presented together with a number of variations. Unlike other classical algorithms such as the Baum-Welch algorithm, the algorithms described are smooth and can be used on-line (after each example presentation) or in batch mode, with or without the usual Viterbi most likely path approximation. The algorithms have simple expressions that result from using a normalized-exponential representation for the HMM parameters. All the algorithms presented are proved to be exact or approximate gradient optimization algorithms with respect to likelihood, log-likelihood, or cross-entropy functions, and as such are usually convergent. These algorithms can also be casted in the more general EM (Expectation-Maximization) framework where they can be viewed as exact or approximate GEM (Generalized Expectation-Maximization) algorithms. The mathematical properties of the algorithms are derived in the appendix.

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23. On Functional Approximation with Normalized Gaussian Units

Michel Bena■■m

https://doi.org/10.1162/neco.1994.6.2.319 | pp. 319-333

Feedforward neural networks with a single hidden layer using normalized gaussian units are studied. It is proved that such neural networks are capable of universal approximation in a satisfactory sense. Then, a hybrid learning rule as per Moody and Darken that combines unsupervised learning of hidden units and supervised learning of output units is considered. By using the method of ordinary differential equations for adaptive algorithms (ODE method) it is shown that the asymptotic properties of the learning rule may be studied in terms of an autonomous cascade of dynamical systems. Some recent results from Hirsch about cascades are used to show the asymptotic stability of the learning rule.

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24. Statistical Physics, Mixtures of Distributions, and the EM Algorithm

Alan Yuille, Paul Stolorz, Joachim Utans

https://doi.org/10.1162/neco.1994.6.2.334 | pp. 334-340

We show that there are strong relationships between approaches to optmization and learning based on statistical physics or mixtures of experts. In particular, the EM algorithm can be interpreted as converging either to a local maximum of the mixtures model or to a saddle point solution to the statistical physics system. An advantage of the statistical physics approach is that it naturally gives rise to a heuristic continuation method, deterministic annealing, for finding good solutions.

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25. Statistical Physics Algorithms That Converge

Alan Yuille, J.J. Kosowsky

https://doi.org/10.1162/neco.1994.6.3.341 | pp. 341-356

In recent years there has been significant interest in adapting techniques from statistical physics, in particular mean field theory, to provide deterministic heuristic algorithms for obtaining approximate solutions to optimization problems. Although these algorithms have been shown experimentally to be successful there has been little theoretical analysis of them. In this paper we demonstrate connections between mean field theory methods and other approaches, in particular, barrier function and interior point methods. As an explicit example, we summarize our work on the linear assignment problem. In this previous work we defined a number of algorithms, including deterministic annealing, for solving the assignment problem. We proved convergence, gave bounds on the convergence times, and showed relations to other optimization algorithms.

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26. Object Recognition and Sensitive Periods: A Computational Analysis of Visual Imprinting

Randall C. O'Reilly, Mark H. Johnson

https://doi.org/10.1162/neco.1994.6.3.357 | pp. 357-389

Using neural and behavioral constraints from a relatively simple biological visual system, we evaluate the mechanism and behavioral implications of a model of invariant object recognition. Evidence from a variety of methods suggests that a localized portion of the domestic chick brain, the intermediate and medial hyperstriatum ventrale (IMHV), is critical for object recognition. We have developed a neural network model

of translation-invariant object recognition that incorporates features of the neural circuitry of IMHV, and exhibits behavior qualitatively similar to a range of findings in the filial imprinting paradigm. We derive several counter-intuitive behavioral predictions that depend critically upon the biologically derived features of the model. In particular, we propose that the recurrent excitatory and lateral inhibitory circuitry in the model, and observed in IMHV, produces hysteresis on the activation state of the units in the model and the principal excitatory neurons in IMHV. Hysteresis, when combined with a simple Hebbian covariance learning mechanism, has been shown in this and earlier work (Földiák 1991; O'Reilly and McClelland 1992) to produce translation-invariant visual representations. The hysteresis and learning rule are responsible for a sensitive period phenomenon in the network, and for a series of novel temporal blending phenomena. These effects are empirically testable. Further, physiological and anatomical features of mammalian visual cortex support a hysteresis-based mechanism, arguing for the generality of the algorithm.

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27. Computing Stereo Disparity and Motion with Known Binocular Cell Properties

Ning Qian

https://doi.org/10.1162/neco.1994.6.3.390 | pp. 390-404

Many models for stereo disparity computation have been proposed, but few can be said to be truly biological. There is also a rich literature devoted to physiological studies of stereopsis. Cells sensitive to binocular disparity have been found in the visual cortex, but it is not clear whether these cells could be used to compute disparity maps from stereograms. Here we propose a model for biological stereo vision based on known receptive field profiles of binocular cells in the visual cortex and provide the first demonstration that these cells could effectively solve random dot stereograms. Our model also allows a natural integration of stereo vision and motion detection. This may help explain the existence of units tuned to both disparity and motion in the visual cortex.

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28. Integration and Differentiation in Dynamic Recurrent Neural Networks

Edwin E. Munro, Larry Shupe, Eberhard E. Fetz

https://doi.org/10.1162/neco.1994.6.3.405 | pp. 405-419

Dynamic neural networks with recurrent connections were trained by backpropagation to generate the differential or the leaky integral of a nonrepeating frequency-modulated sinusoidal signal. The trained networks performed these operations on arbitrary input waveforms. Reducing the network size by deleting ineffective hidden units and combining redundant units, and then retraining the network produced a minimal network that computed the same function and revealed the underlying computational algorithm. Networks could also be trained to compute simultaneously the differential and integral of the input on two outputs; the two operations were performed in distributed overlapping fashion, and the activations of the hidden units were dominated by the integral. Incorporating units with time constants into model networks generally enhanced their performance as integrators and interfered with their ability to differentiate.

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29. A Convergence Result for Learning in Recurrent Neural Networks

Chung■Ming Kuan, Kurt Hornik, Halbert White
https://doi.org/10.1162/neco.1994.6.3.420 | pp. 420-440

We give a rigorous analysis of the convergence properties of a backpropagation algorithm for recurrent networks containing either output or hidden layer recurrence. The conditions permit data generated by stochastic processes with considerable dependence. Restrictions are offered that may help assure convergence of the network parameters to a local optimum, as some simulations illustrate.

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30. Topology Learning Solved by Extended Objects: A Neural Network Model

Csaba Szepesvári, László Balázs, András Lörincz
https://doi.org/10.1162/neco.1994.6.3.441 | pp. 441-458

It is shown that local, extended objects of a metrical topological space shape the receptive fields of competitive neurons to local filters. Self-organized topology learning is then solved with the help of Hebbian learning together with extended objects that provide unique information about neighborhood relations. A topographical map is deduced and is used to speed up further adaptation in a changing environment with the help of Kohonen-type learning that teaches the neighbors of winning neurons as well.

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31. Dynamics of Discrete Time, Continuous State Hopfield Networks

Pascal Koiran
https://doi.org/10.1162/neco.1994.6.3.459 | pp. 459-468

The dynamics of discrete time, continuous state Hopfield networks is driven by an energy function. In this paper, we use this tool to prove under mild hypotheses that any trajectory converges to a fixed point for the sequential iteration, and to a cycle of length 2 or a fixed point for the parallel iteration. Perhaps surprisingly, it seems that no rigorous proof of these results was published before.

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32. Alopex: A Correlation-Based Learning Algorithm for Feedforward and Recurrent Neural Networks

K. P. Unnikrishnan, K. Venugopal
https://doi.org/10.1162/neco.1994.6.3.469 | pp. 469-490

We present a learning algorithm for neural networks, called Alopex. Instead of error gradient, Alopex uses local correlations between changes in individual weights and changes in the global error measure. The algorithm does not make any assumptions about transfer functions of individual neurons, and does not explicitly depend on the functional form of the error measure. Hence, it can be used in networks with arbitrary transfer functions and for minimizing a large class of error measures. The learning algorithm is the same for feedforward and recurrent networks. All the weights in a network are updated simultaneously, using only local computations. This allows complete parallelization of the algorithm. The algorithm is stochastic and it uses a “temperature” parameter in a manner similar to that in simulated annealing. A heuristic “annealing schedule” is presented that is effective in finding global minima of error surfaces. In this paper, we report extensive simulation studies illustrating these advantages and show that learning times are

comparable to those for standard gradient descent methods. Feedforward networks trained with Alopex are used to solve the MONK's problems and symmetry problems. Recurrent networks trained with the same algorithm are used for solving temporal XOR problems. Scaling properties of the algorithm are demonstrated using encoder problems of different sizes and advantages of appropriate error measures are illustrated using a variety of problems.

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33. Duality Between Learning Machines: A Bridge Between Supervised and Unsupervised Learning

Jean■Pierre Nadal, Néstor Parga

https://doi.org/10.1162/neco.1994.6.3.491 | pp. 491-508

We exhibit a duality between two perceptrons that allows us to compare the theoretical analysis of supervised and unsupervised learning tasks. The first perceptron has one output and is asked to learn a classification of p patterns. The second (dual) perceptron has p outputs and is asked to transmit as much information as possible on a distribution of inputs. We show in particular that the maximum information that can be stored in the couplings for the supervised learning task is equal to the maximum information that can be transmitted by the dual perceptron.

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34. Finding the Embedding Dimension and Variable Dependencies in Time Series

Hong Pi, Carsten Peterson

https://doi.org/10.1162/neco.1994.6.3.509 | pp. 509-520

We present a general method, the δ-test, which establishes functional dependencies given a sequence of measurements. The approach is based on calculating conditional probabilities from vector component distances. Imposing the requirement of continuity of the underlying function, the obtained values of the conditional probabilities carry information on the embedding dimension and variable dependencies. The power of the method is illustrated on synthetic time-series with different time-lag dependencies and noise levels and on the sunspot data. The virtue of the method for preprocessing data in the context of feedforward neural networks is demonstrated. Also, its applicability for tracking residual errors in output units is stressed.

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35. Comparison of Some Neural Network and Scattered Data Approximations: The Inverse Manipulator Kinematics Example

Dimitry Gorinevsky, Thomas H. Connolly

https://doi.org/10.1162/neco.1994.6.3.521 | pp. 521-542

This paper compares the application of five different methods for the approximation of the inverse kinematics of a manipulator arm from a number of joint angle/Cartesian coordinate training pairs. The first method is a standard feedforward neural network with error backpropagation learning. The next two methods are derived from an extended Kohonen Map algorithm that we combine with Shepard interpolation for the forward computation. We compare the method of Ritter et al. for the learning of the extended

Kohonen Map to our own scheme based on gradient descent optimization. We also study three scattered data approximation algorithms. They include two variants of the Radial Basis Function (RBF) method: Hardy's multiquadrics and gaussian RBF. We further develop our own Local Polynomial Fit method that could be considered as a modification of McLain's method. We propose extensions to the considered scattered data approximation algorithms to make them suitable for vector-valued multivariable functions, such as the mapping of Cartesian coordinates into joint angle coordinates.

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36. Functionally Equivalent Feedforward Neural Networks

V■ra K■rková, Paul C. Kainen

https://doi.org/10.1162/neco.1994.6.3.543 | pp. 543-558

For a feedforward perceptron type architecture with a single hidden layer but with a quite general activation function, we characterize the relation between pairs of weight vectors determining networks with the same input-output function.

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37. What Is the Goal of Sensory Coding?

David J. Field

https://doi.org/10.1162/neco.1994.6.4.559 | pp. 559-601

A number of recent attempts have been made to describe early sensory coding in terms of a general information processing strategy. In this paper, two strategies are contrasted. Both strategies take advantage of the redundancy in the environment to produce more effective representations. The first is described as a “compact” coding scheme. A compact code performs a transform that allows the input to be represented with a reduced number of vectors (cells) with minimal RMS error. This approach has recently become popular in the neural network literature and is related to a process called Principal Components Analysis (PCA). A number of recent papers have suggested that the optimal “compact” code for representing natural scenes will have units with receptive field profiles much like those found in the retina and primary visual cortex. However, in this paper, it is proposed that compact coding schemes are insufficient to account for the receptive field properties of cells in the mammalian visual pathway. In contrast, it is proposed that the visual system is near to optimal in representing natural scenes only if optimality is defined in terms of “sparse distributed” coding. In a sparse distributed code, all cells in the code have an equal response probability across the class of images but have a low response probability for any single image. In such a code, the dimensionality is not reduced. Rather, the redundancy of the input is transformed into the redundancy of the firing pattern of cells. It is proposed that the signature for a sparse code is found in the fourth moment of the response distribution (i.e., the kurtosis). In measurements with 55 calibrated natural scenes, the kurtosis was found to peak when the bandwidths of the visual code matched those of cells in the mammalian visual cortex. Codes resembling “wavelet transforms” are proposed to be effective because the response histograms of such codes are sparse (i.e., show high kurtosis) when presented with natural scenes. It is proposed that the structure of the image that allows sparse coding is found in the phase spectrum of the image. It is suggested that natural scenes, to a first approximation, can be considered as a sum of self-similar local functions (the inverse of a wavelet). Possible reasons for why sensory systems would evolve toward sparse coding are presented.

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38. Design Principles of Columnar Organization in Visual Cortex

Ernst Niebur, Florentin Wörgötter

https://doi.org/10.1162/neco.1994.6.4.602 | pp. 602-614

Visual space is represented by cortical cells in an orderly manner. Only little variation in the cell behavior is found with changing depth below the cortical surface, that is, all cells in a column with axis perpendicular to the cortical plane have approximately the same properties (Hubel and Wiesel 1962, 1963, 1968). Therefore, the multiple features of the visual space (e.g., position in visual space, preferred orientation, and orientation tuning strength) are mapped on a two-dimensional space, the cortical plane. Such a dimension reduction leads to complex maps (Durbin and Mitchison 1990) that so far have evaded an intuitive understanding. Analyzing optical imaging data (Blasdel 1992a, b; Blasdel and Salama 1986; Grinvald et al. 1986) using a theoretical approach we will show that the most salient features of these maps can be understood from a few basic design principles: local correlation, modularity, isotropy, and homogeneity. These principles can be defined in a mathematically exact sense in the Fourier domain by a rather simple annulus-like spectral structure. Many of the models that have been developed to explain the mapping of the preferred orientations (Cooper et al. 1979; Legendy 1978; Linsker 1986a, b; Miller 1992; Nass and Cooper 1975; Obermayer et al. 1990, 1992; Soodak 1987; Swindale 1982, 1985, 1992; von der Malsburg 1973; von der Malsburg and Cowan 1982) are quite successful in generating maps that are close to experimental maps. We suggest that this success is due to these principles, which are common properties of the models and of biological maps.

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39. Elastic Net Model of Ocular Dominance: Overall Stripe Pattern and Monocular Deprivation

Geoffrey J. Goodhill, David Willshaw

https://doi.org/10.1162/neco.1994.6.4.615 | pp. 615-621

The elastic net (Durbin and Willshaw 1987) can account for the development of both topography and ocular dominance in the mapping from the lateral geniculate nucleus to primary visual cortex (Goodhill and Willshaw 1990). Here it is further shown for this model that (1) the overall pattern of stripes produced is strongly influenced by the shape of the cortex: in particular, stripes with a global order similar to that seen biologically can be produced under appropriate conditions, and (2) the observed changes in stripe width associated with monocular deprivation are reproduced in the model.

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40. The Effect of Synchronized Inputs at the Single Neuron Level

Öjvind Bernander, Christof Koch, Marius Usher

https://doi.org/10.1162/neco.1994.6.4.622 | pp. 622-641

It is commonly assumed that temporal synchronization of excitatory synaptic inputs onto a single neuron increases its firing rate. We investigate here the role of synaptic synchronization for the leaky integrate-and-fire neuron as well as for a biophysically and anatomically detailed compartmental model of a cortical pyramidal cell. We find that if the number of excitatory inputs, N, is on the same order as the number of fully synchronized inputs necessary to trigger a single action potential, N t , synchronization always increases the firing rate (for both constant and Poisson-distributed input). However, for large values of N compared to N t , “overcrowding” occurs and temporal synchronization is detrimental to firing frequency.

This behavior is caused by the conflicting influence of the low-pass nature of the passive dendritic membrane on the one hand and the refractory period on the other. If both temporal synchronization as well as the fraction of synchronized inputs (Murthy and Fetz 1993) is varied, synchronization is only advantageous if either N or the average input frequency, f in , are small enough.

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41. Segmentation by a Network of Oscillators with Stored Memories

Haim Sompolinsky, Misha Tsodyks

https://doi.org/10.1162/neco.1994.6.4.642 | pp. 642-657

We propose a model of coupled oscillators with noise that performs segmentation of stimuli using a set of stored images, each consisting of objects and a background. The oscillators' amplitudes encode the spatial and featural distribution of the external stimulus. The coherence of their phases signifies their belonging to the same object. In the learning stage, the couplings between phases are modified in a Hebb-like manner. By mean-field analysis and simulations, we show that an external stimulus whose local features resemble those of one or several of the stored objects generates a selective phase coherence that represents the stored pattern of segmentation.

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42. Numerical Bifurcation Analysis of an Oscillatory Neural Network with Synchronous/Asynchronous Connections

Yukio Hayashi

https://doi.org/10.1162/neco.1994.6.4.658 | pp. 658-667

One of the advantages of oscillatory neural networks is their dynamic links among related features; the links are based on input-dependent synchronized oscillations. This paper investigates the relations between synchronous/asynchronous oscillations and the connection architectures of an oscillatory neural network with two excitatory-inhibitory pair oscillators. Through numerical analysis, we show synchronous and asynchronous connection types over a wide parameter space for two different inputs and one connection parameter. The results are not only consistent with the classification of synchronous/asynchronous connection types in König's model (1991), but also offer a useful guideline on how to construct a network with local connections for a segmentation task.

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43. Analysis of the Effects of Noise on a Model for the Neural Mechanism of Short-Term Active Memory

J. Devin McAuley, Joseph G. Stampfli

https://doi.org/10.1162/neco.1994.6.4.668 | pp. 668-678

Zipser (1991) showed that the hidden unit activity of a fully recurrent neural network model, trained on a simple memory task, matched the temporal activity patterns of memory-associated neurons in monkeys performing delayed saccade or delayed match-to-sample tasks. When noise, simulating random fluctuations in neural firing rate, is added to the unit activations of this model, the effect on the memory dynamics is to slow the rate of information loss. In this paper, we show that the dynamics of the iterated sigmoid function, with gain and bias parameters, is qualitatively very similar to the tonic response properties of Zipser's

multiunit model. Analysis of the simpler system provides an explanation for the effect of noise that is missing from the description of the multiunit model.

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44. Reduction of Conductance-Based Models with Slow Synapses to Neural Nets

Bard Ermentrout

https://doi.org/10.1162/neco.1994.6.4.679 | pp. 679-695

The method of averaging and a detailed bifurcation calculation are used to reduce a system of synaptically coupled neurons to a Hopfield type continuous time neural network. Due to some special properties of the bifurcation, explicit averaging is not required and the reduction becomes a simple algebraic problem. The resultant calculations show one how to derive a new type of “squashing function” whose properties are directly related to the detailed ionic mechanisms of the membrane. Frequency encoding as opposed to amplitude encoding emerges in a natural fashion from the theory. The full system and the reduced system are numerically compared.

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45. Dimension Reduction of Biological Neuron Models by Artificial Neural Networks

Kenji Doya, Allen I. Selverston

https://doi.org/10.1162/neco.1994.6.4.696 | pp. 696-717

An artificial neural network approach to dimension reduction of dynamical systems is proposed and applied to conductance-based neuron models. Networks with bottleneck layers of continuous-time dynamical units could make a two-dimensional model from the trajectories of the Hodgkin-Huxley model and a three-dimensional model from the trajectories of a six-dimensional bursting neuron model. Nullcline analysis of these reduced models revealed the bifurcations of the dynamical system underlying firing and bursting behaviors.

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46. Neural Network Process Models Based on Linear Model Structures

Gary M. Scott, W. Harmon Ray

https://doi.org/10.1162/neco.1994.6.4.718 | pp. 718-738

The KBANN (Knowledge-Based Artificial Neural Networks) approach uses neural networks to refine knowledge that can be written in the form of simple propositional rules. This idea is extended by presenting the MANNIDENT (Multivariable Artificial Neural Network Identification) algorithm by which the mathematical equations of linear dynamic process models determine the topology and initial weights of a network, which is further trained using backpropagation. This method is applied to the task of modeling a nonisothermal chemical reactor in which a first-order exothermic reaction is occurring. This method produces statistically significant gains in accuracy over both a standard neural network approach and a linear model. Furthermore, using the approximate linear model to initialize the weights of the network produces statistically less variation in model fidelity. By structuring the neural network according to the approximate linear model, the model can be readily interpreted.

47. Stability of Oja's PCA Subspace Rule

Juha Karhunen

https://doi.org/10.1162/neco.1994.6.4.739 | pp. 739-747

This paper deals with stability of Oja's symmetric algorithm for estimating the principal component subspace of the input data. Exact conditions are derived for the gain parameter on which the discrete algorithm remains bounded. The result is extended for a nonlinear version of Oja's algorithm.

48. Supervised Training of Neural Networks via Ellipsoid Algorithms

Man-Fung Cheung, Kevin M. Passino, Stephen Yurkovich

https://doi.org/10.1162/neco.1994.6.4.748 | pp. 748-760

In this paper we show that two ellipsoid algorithms can be used to train single-layer neural networks with general staircase nonlinearities. The ellipsoid algorithms have several advantages over other conventional training approaches including (1) explicit convergence results and automatic determination of linear separability, (2) an elimination of problems with picking initial values for the weights, (3) guarantees that the trained weights are in some “acceptable region,” (4) certain “robustness” characteristics, and (5) a training approach for neural networks with a wider variety of activation functions. We illustrate the training approach by training the MAJ function and then by showing how to train a controller for a reaction chamber temperature control problem.

49. Why Some Feedforward Networks Cannot Learn Some Polynomials

N. Scott Cardell, Wayne Joerding, Ying Li

https://doi.org/10.1162/neco.1994.6.4.761 | pp. 761-766

It seems natural to test feedforward networks on deterministic functions. Yet, some simple functions, notably polynomials, present some difficult problems for approximation by feedforward networks. The estimated parameters become unbounded and fail to follow any unique pattern. Furthermore, as the fit to the specified functions becomes closer, numerical problems may develop in the algorithm. This paper explains why these problems occur for polynomials of order less than or equal to the number of hidden units of a feedforward network. We show that other examples occur for functions mathematically related to the network's squashing function. These difficulties do not indicate problems with the training algorithm, but occur as an inherent consequence of the role of the connection weights in feedforward networks.

50. A Bayesian Analysis of Self-Organizing Maps

S.P. Luttrell

https://doi.org/10.1162/neco.1994.6.5.767 | pp. 767-794

In this paper Bayesian methods are used to analyze some of the properties of a special type of Markov chain. The forward transitions through the chain are followed by inverse transitions (using Bayes' theorem) backward through a copy of the same chain; this will be called a folded Markov chain. If an appropriately

defined Euclidean error (between the original input and its “reconstruction” via Bayes' theorem) is minimized with respect to the choice of Markov chain transition probabilities, then the familiar theories of both vector quantizers and self-organizing maps emerge. This approach is also used to derive the theory of self-supervision, in which the higher layers of a multilayer network supervise the lower layers, even though overall there is no external teacher.

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51. Network Amplification of Local Fluctuations Causes High Spike Rate Variability, Fractal Firing Patterns and Oscillatory Local Field Potentials

Marius Usher, Martin Stemmler, Christof Koch, Zeev Olami

https://doi.org/10.1162/neco.1994.6.5.795 | pp. 795-836

We investigate a model for neural activity in a two-dimensional sheet of leaky integrate-and-fire neurons with feedback connectivity consisting of local excitation and surround inhibition. Each neuron receives stochastic input from an external source, independent in space and time. As recently suggested by Softky and Koch (1992, 1993), independent stochastic input alone cannot explain the high interspike interval variability exhibited by cortical neurons in behaving monkeys. We show that high variability can be obtained due to the amplification of correlated fluctuations in a recurrent network. Furthermore, the cross-correlation functions have a dual structure, with a sharp peak on top of a much broader hill. This is due to the inhibitory and excitatory feedback connections, which cause “hotspots” of neural activity to form within the network. These localized patterns of excitation appear as clusters or stripes that coalesce, disintegrate, or fluctuate in size while simultaneously moving in a random walk constrained by the interaction with other clusters. The synaptic current impinging upon a single neuron shows large fluctuations at many time scales, leading to a large coefficient of variation (C V) for the interspike interval statistics. The power spectrum associated with single units shows a 1/f decay for small frequencies and is flat at higher frequencies, while the power spectrum of the spiking activity averaged over many cells—equivalent to the local field potential—shows no 1/f decay but a prominent peak around 40 Hz, in agreement with data recorded from cat and monkey cortex (Gray et al. 1990; Eckhorn et al. 1993). Firing rates exhibit self-similarity between 20 and 800 msec, resulting in 1/f-like noise, consistent with the fractal nature of neural spike trains (Teich 1992).

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52. Statistical Analysis of an Autoassociative Memory Network

Alan M.N. Fu

https://doi.org/10.1162/neco.1994.6.5.837 | pp. 837-841

A statistical method for analyzing the dynamic behavior of a synchronous autoassociative memory network is presented based on an extension of Amari-Maginu theory. Through computer simulations it is shown that when the memorized pattern ratio is small (r < 0.15) or large (r > 0.2), the derived theory shows good agreement.

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53. Loading Deep Networks Is Hard

Ji■■■ Š■■ma

https://doi.org/10.1162/neco.1994.6.5.842 | pp. 842-850

The loading problem formulated by J. S. Judd seems to be a relevant model for supervised connectionist learning of the feedforward networks from the complexity point of view. It is known that loading general network architectures is NP-complete (intractable) when the (training) tasks are also general. Many strong restrictions on architectural design and/or on the tasks do not help to avoid the intractability of loading. Judd concentrated on the width expanding architectures with constant depth and found a polynomial time algorithm for loading restricted shallow architectures. He suppressed the effect of depth on loading complexity and left as an open prototypical computational problem the loading of easy regular triangular architectures that might capture the crux of depth difficulties. We have proven this problem to be NP-complete. This result does not give much hope for the existence of an efficient algorithm for loading deep networks.

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54. Measuring the VC-Dimension of a Learning Machine

Vladimir Vapnik, Esther Levin, Y. Le Cun

https://doi.org/10.1162/neco.1994.6.5.851 | pp. 851-876

A method for measuring the capacity of learning machines is described. The method is based on fitting a theoretically derived function to empirical measurements of the maximal difference between the error rates on two separate data sets of varying sizes. Experimental measurements of the capacity of various types of linear classifiers are presented.

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55. Neural Nets with Superlinear VC-Dimension

Wolfgang Maass

https://doi.org/10.1162/neco.1994.6.5.877 | pp. 877-884

It has been known for quite a while that the Vapnik-Chervonenkis dimension (VC-dimension) of a feedforward neural net with linear threshold gates is at most $O(w \cdot \log w)$, where w is the total number of weights in the neural net. We show in this paper that this bound is in fact asymptotically optimal. More precisely, we exhibit for any depth $d \geq 3$ a large class of feedforward neural nets of depth d with w weights that have VC-dimension $\Omega(w \cdot \log w)$. This lower bound holds even if the inputs are restricted to Boolean values. The proof of this result relies on a new method that allows us to encode more “program-bits” in the weights of a neural net than previously thought possible.

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56. A Novel Design Method for Multilayer Feedforward Neural Networks

Jihong Lee

https://doi.org/10.1162/neco.1994.6.5.885 | pp. 885-901

A multilayer feedforward neural network and a design method that takes the distribution of given training patterns into consideration are proposed. The size of the network, initial values of interconnection weights, and parameters defining the nonlinearities of processing elements are determined from a specially selected portion of the given training patterns, which is called a set of feature points. With these initial settings, the performance of the network is further improved by a modified error backpropagation learning process. It is shown in several examples that the proposed model and the design method are capable of rapidly learning

the training patterns compared to conventional multilayer feedforward neural networks with random initialization techniques.

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57. An Internal Mechanism for Detecting Parasite Attractors in a Hopfield Network

Jean■Dominique Gascuel, Bahram Moobed, M. Weinfeld

https://doi.org/10.1162/neco.1994.6.5.902 | pp. 902-915

This paper presents a built-in mechanism for automatic detection of prototypes (as opposed to parasite attractors) in a Hopfield network. It has a good statistical performance and avoids host computer overhead for this purpose. This mechanism is based on an internal coding of the prototypes during learning, using cyclic redundancy codes, and leads to an efficient implementation in VLSI. The immediate separation of prototypes from parasite attractors can be used to enhance the autonomy of the networks, aiming at hierarchical multinetwork structures. As an example, the use of such an architecture for the classification of handwritten digits is described.

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58. On Langevin Updating in Multilayer Perceptrons

Thorsteinn Rögnvaldsson

https://doi.org/10.1162/neco.1994.6.5.916 | pp. 916-926

The Langevin updating rule, in which noise is added to the weights during learning, is presented and shown to improve learning on problems with initially ill-conditioned Hessians. This is particularly important for multilayer perceptrons with many hidden layers, that often have ill-conditioned Hessians. In addition, Manhattan updating is shown to have a similar effect.

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59. Probabilistic Winner-Take-All Learning Algorithm for Radial-Basis-Function Neural Classifiers

Hossam Osman, M.M. Fahmy

https://doi.org/10.1162/neco.1994.6.5.927 | pp. 927-943

This paper proposes a new adaptive competitive learning algorithm called “the probabilistic winner-take-all.” The algorithm is based on a learning scheme developed by Agrawala within the statistical pattern recognition literature (Agrawala 1970). Its name stems from the fact that for a given input pattern once each competitor computes the probability of being the one that generated this pattern, the computed probabilities are utilized to probabilistically choose a winner. Then, only this winner is permitted to learn. The learning rule of the algorithm is derived for three different cases. Its properties are discussed and compared to those of two other competitive learning algorithms, namely the standard winner-take-all and the maximum-likelihood soft competition. Experimental comparison is also given. When all three algorithms are used to train the hidden layer of radial-basis-function classifiers, experiments indicate that classifiers trained with the probabilistic winner-take-all outperform those trained with the other two algorithms.

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60. Realization of the “Weak Rod” by a Double Layer Parallel Network

Takashi Matsumoto, K. Kondo

https://doi.org/10.1162/neco.1994.6.5.944 | pp. 944-956

The weak rod can be realized by a very natural double layer network with feedback. While the second layer is linear, the first layer is significantly nonlinear. Elements associated with the first layer are called “synaptic fuses.” An MRF argument is used to eliminate the line variables. The network makes what is cut transparent when a crease or a step edge is present.

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61. Learning in Neural Networks with Material Synapses

Daniel J. Amit, Stefano Fusi

https://doi.org/10.1162/neco.1994.6.5.957 | pp. 957-982

We discuss the long term maintenance of acquired memory in synaptic connections of a perpetually learning electronic device. This is affected by ascribing each synapse a finite number of stable states in which it can maintain for indefinitely long periods. Learning uncorrelated stimuli is expressed as a stochastic process produced by the neural activities on the synapses. In several interesting cases the stochastic process can be analyzed in detail, leading to a clarification of the performance of the network, as an associative memory, during the process of uninterrupted learning. The stochastic nature of the process and the existence of an asymptotic distribution for the synaptic values in the network imply generically that the memory is a palimpsest but capacity is as low as log N for a network of N neurons. The only way we find for avoiding this tight constraint is to allow the parameters governing the learning process (the coding level of the stimuli; the transition probabilities for potentiation and depression and the number of stable synaptic levels) to depend on the number of neurons. It is shown that a network with synapses that have two stable states can dynamically learn with optimal storage efficiency, be a palimpsest, and maintain its (associative) memory for an indefinitely long time provided the coding level is low and depression is equilibrated against potentiation. We suggest that an option so easily implementable in material devices would not have been overlooked by biology. Finally we discuss the stochastic learning on synapses with variable number of stable synaptic states.

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62. Model Based on Extracellular Potassium for Spontaneous Synchronous Activity in Developing Retinas

Pierre■Yves Burgi, Norberto M. Grzywacz

https://doi.org/10.1162/neco.1994.6.5.983 | pp. 983-1004

Waves of action-potential bursts propagate across the ganglion-cell surface of isolated developing retinas. It has been suggested that the rise of extracellular potassium concentration following a burst of action potentials in a cell may underlie these waves by depolarizing neighbor cells. This suggestion is sensible for developing tissues, since their glial system is immature. We tested whether this extracellular-potassium suggestion is feasible. For this purpose, we built a realistic biophysical model of the ganglion-cell layer of the developing retina. Simulations with this model show that increases of extracellular potassium are sufficiently high (about fourfold) to mediate the waves consistently with experimental physiological and pharmacological data. Even if another mechanism mediates the waves, these simulations indicate that extracellular

potassium should significantly modulate the waves' properties.

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63. Bayesian Modeling and Classification of Neural Signals

Michael S. Lewicki

https://doi.org/10.1162/neco.1994.6.5.1005 | pp. 1005-1030

Identifying and classifying action potential shapes in extracellular neural waveforms have long been the subject of research, and although several algorithms for this purpose have been successfully applied, their use has been limited by some outstanding problems. The first is how to determine shapes of the action potentials in the waveform and, second, how to decide how many shapes are distinct. A harder problem is that action potentials frequently overlap making difficult both the determination of the shapes and the classification of the spikes. In this report, a solution to each of these problems is obtained by applying Bayesian probability theory. By defining a probabilistic model of the waveform, the probability of both the form and number of spike shapes can be quantified. In addition, this framework is used to obtain an efficient algorithm for the decomposition of arbitrarily complex overlap sequences. This algorithm can extract many times more information than previous methods and facilitates the extracellular investigation of neuronal classes and of interactions within neuronal circuits.

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64. Information Processing in Dendritic Trees

Bartlett W. Mel

https://doi.org/10.1162/neco.1994.6.6.1031 | pp. 1031-1085

This review considers the input-output behavior of neurons with dendritic trees, with an emphasis on questions of information processing. The parts of this review are (1) a brief history of ideas about dendritic trees, (2) a review of the complex electrophysiology of dendritic neurons, (3) an overview of conceptual tools used in dendritic modeling studies, including the cable equation and compartmental modeling techniques, and (4) a review of modeling studies that have addressed various issues relevant to dendritic information processing.

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65. Simulations of Intrinsically Bursting Neocortical Pyramidal Neurons

Paul Rhodes, Charles M. Gray

https://doi.org/10.1162/neco.1994.6.6.1086 | pp. 1086-1110

Neocortical layer 5 intrinsically bursting (IB) pyramidal neurons were simulated using compartment model methods. Morphological data as well as target neurophysiological responses were taken from a series of published studies on the same set of rat visual cortex pyramidal neurons (Mason, A. and Larkman, A. J., 1990. J. Neurosci. 9,1440-1447; Larkman, A. J. 1991. J. Comp. Neurol. 306, 307-319). A dendritic distribution of ion channels was found that reproduced the range of in vitro responses of layer 5 IB pyramidal neurons, including the transition from repetitive bursting to the burst/tonic spiking mode seen in these neurons as input magnitude increases. In light of available data, the simulation results suggest that in these neurons bursts are driven by an inward flow of current during a high threshold Ca 2+ spike extending throughout both the basal and apical dendritic branches.

66. Effects of Input Synchrony on the Firing Rate of a Three-Conductance Cortical Neuron Model

Venkatesh N. Murthy, Eberhard E. Fetz

https://doi.org/10.1162/neco.1994.6.6.1111 | pp. 1111-1126

For a model cortical neuron with three active conductances, we studied the dependence of the firing rate on the degree of synchrony in its synaptic inputs. The effect of synchrony was determined as a function of three parameters: number of inputs, average input frequency, and the synaptic strength (maximal unitary conductance change). Synchrony alone could increase the cell's firing rate when the product of these three parameters was below a critical value. But for higher values of the three parameters, synchrony could reduce firing rate. Instantaneous responses to time-varying input firing rates were close to predictions from steady-state responses when input synchrony was high, but fell below steady-state responses when input synchrony was low. Effectiveness of synaptic transmission, measured by the peak area of cross-correlations between input and output spikes, increased with increasing synchrony.

67. The Functional Role of Excitatory and Inhibitory Interactions in Chopper Cells of the Anteroventral Cochlear Nucleus

Ying-Cheng Lai, Raimond L. Winslow, Murray B. Sachs

https://doi.org/10.1162/neco.1994.6.6.1127 | pp. 1127-1140

Chopper cells in the anteroventral cochlear nucleus of the cat maintain a robust rate-place representation of vowel spectra over a broad range of stimulus levels. This representation resembles that of low threshold, high spontaneous rate primary auditory nerve fibers at low stimulus levels, and that of high threshold, low spontaneous rate auditory-nerve fibers at high stimulus levels. This has led to the hypothesis that chopper cells in the anteroventral cochlear nucleus selectively process inputs from different spontaneous rate populations of primary auditory-nerve fibers at different stimulus levels. We present a computational model, making use of shunting inhibition, for how this level dependent processing may be performed within the chopper cell dendritic tree. We show that this model (1) implements level-dependent selective processing, (2) reproduces detailed features of real chopper cell post-stimulus-time histograms, and (3) reproduces nonmonotonic rate versus level functions in response to single tones measured.

68. Control of Chaos in Networks with Delay: A Model for Synchronization of Cortical Tissue

Carlos Lourenço, A. Babloyantz

https://doi.org/10.1162/neco.1994.6.6.1141 | pp. 1141-1154

The unstable periodic orbits of chaotic dynamics in systems described by delay differential equations are considered. An orbit is stabilized successfully, using a method proposed by Pyragas. The system under investigation is a network of excitatory and inhibitory neurons of moderate size, describing cortical activity. The relevance of the results for synchronized cortical activity is discussed.

69. First-Order Recurrent Neural Networks and Deterministic Finite State Automata

Peter Manolios, Robert Fanelli

https://doi.org/10.1162/neco.1994.6.6.1155 | pp. 1155-1173

We examine the correspondence between first-order recurrent neural networks and deterministic finite state automata. We begin with the problem of inducing deterministic finite state automata from finite training sets, that include both positive and negative examples, an NP-hard problem (Angluin and Smith 1983). We use a neural network architecture with two recurrent layers, which we argue can approximate any discrete-time, time-invariant dynamic system, with computation of the full gradient during learning. The networks are trained to classify strings as belonging or not belonging to the grammar. The training sets used contain only short strings, and the sets are constructed in a way that does not require a priori knowledge of the grammar. After training, the networks are tested using various test sets with strings of length up to 1000, and are often able to correctly classify all the test strings. These results are comparable to those obtained with second-order networks (Giles et al. 1992; Watrous and Kuhn 1992a; Zeng et al. 1993). We observe that the networks emulate finite state automata, confirming the results of other authors, and we use a vector quantization algorithm to extract deterministic finite state automata after training and during testing of the networks, obtaining a table listing the start state, accept states, reject states, all transitions from the states, as well as some useful statistics. We examine the correspondence between finite state automata and neural networks in detail, showing two major stages in the learning process. To this end, we use a graphics module, which graphically depicts the states of the network during the learning and testing phases. We examine the networks' performance when tested on strings much longer than those in the training set, noting a measure based on clustering that is correlated to the stability of the networks. Finally, we observe that with sufficiently long training times, neural networks can become true finite state automata, due to the attractor structure of their dynamics.

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70. Learning in Boltzmann Trees

Lawrence K. Saul, Michael I. Jordan

https://doi.org/10.1162/neco.1994.6.6.1174 | pp. 1174-1184

We introduce a large family of Boltzmann machines that can be trained by standard gradient descent. The networks can have one or more layers of hidden units, with tree-like connectivity. We show how to implement the supervised learning algorithm for these Boltzmann machines exactly, without resort to simulated or mean-field annealing. The stochastic averages that yield the gradients in weight space are computed by the technique of decimation. We present results on the problems of N-bit parity and the detection of hidden symmetries.

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71. On the Convergence of Stochastic Iterative Dynamic Programming Algorithms

Tommi Jaakkola, Michael I. Jordan, Satinder Singh

https://doi.org/10.1162/neco.1994.6.6.1185 | pp. 1185-1201

Recent developments in the area of reinforcement learning have yielded a number of new algorithms for the prediction and control of Markovian environments. These algorithms, including the TD(λ) algorithm of Sutton (1988) and the Q-learning algorithm of Watkins (1989), can be motivated heuristically as approximations to dynamic programming (DP). In this paper we provide a rigorous proof of convergence of these DP-based learning algorithms by relating them to the powerful techniques of stochastic approximation theory via a new convergence theorem. The theorem establishes a general class of convergent algorithms to which both TD(λ) and Q-learning belong.

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72. A Dynamic Neural Network Architecture by Sequential Partitioning of the Input Space

R.S. Shadafan, Mahesan Niranjan

https://doi.org/10.1162/neco.1994.6.6.1202 | pp. 1202-1222

We present a sequential approach to training multilayer perceptrons for pattern classification applications. The network is presented with each item of data only once and its architecture is dynamically adjusted during training. At the arrival of each example, a decision whether to increase the complexity of the network, or simply train the existing nodes is made based on three heuristic criteria. These criteria measure the position of the new item of data in the input space with respect to the information currently stored in the network. During the training process, each layer is assumed to be an independent entity with its particular input space. By adding nodes to each layer, the algorithm is effectively adding a hyperplane to the input space, hence adding a partition in the input space for that layer. When existing nodes are sufficient to accommodate the incoming input, the corresponding hidden nodes will be trained accordingly. Each hidden unit in the network is trained in closed form by means of a recursive least-squares (RLS) algorithm. A local covariance matrix of the data is maintained at each node and the closed form solution is recursively updated. The three criteria are computed from these covariance matrices to keep low computational cost. The performance of the algorithm is illustrated on two problems. The first problem is the two-dimensional Peterson and Barney vowel data. The second problem is a 33-dimensional data derived from a vision system for classifying wheat grains. The sequential nature of the algorithm has an efficient hardware implementation in the form of systolic arrays, and the incremental training idea has better biological plausibility compared with iterative methods.

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73. Pruning from Adaptive Regularization

Lars Kai Hansen, Carl Edward Rasmussen

https://doi.org/10.1162/neco.1994.6.6.1223 | pp. 1223-1232

Inspired by the recent upsurge of interest in Bayesian methods we consider adaptive regularization. A generalization based scheme for adaptation of regularization parameters is introduced and compared to Bayesian regularization. We show that pruning arises naturally within both adaptive regularization schemes. As model example we have chosen the simplest possible: estimating the mean of a random variable with known variance. Marked similarities are found between the two methods in that they both involve a “noise limit,” below which they regularize with infinite weight decay, i.e., they prune. However, pruning is not always beneficial. We show explicitly that both methods in some cases may increase the generalization error. This corresponds to situations where the underlying assumptions of the regularizer are poorly matched to the

environment.

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74. Approximation Capability of Layered Neural Networks with Sigmoid Units on Two Layers

Yoshifusa Ito

https://doi.org/10.1162/neco.1994.6.6.1233 | pp. 1233-1243

Using only an elementary constructive method, we prove the universal approximation capability of three-layered feedforward neural networks that have sigmoid units on two layers. We regard the Heaviside function as a special case of sigmoid function and measure accuracy of approximation in either the supremum norm or in the L p -norm. Given a continuous function defined on a unit hypercube and the required accuracy of approximation, we can estimate the numbers of necessary units on the respective sigmoid unit layers. In the case where the sigmoid function is the Heaviside function, our result improves the estimation of K■rková (1992). If the accuracy of approximation is measured in the L P -norm, our estimation also improves that of K■rková (1992), even when the sigmoid function is not the Heaviside function.

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75. Estimation of Network Parameters in Semiparametric Stochastic Perceptron

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https://doi.org/10.1162/neco.1994.6.6.1244 | pp. 1244-1261

It was reported (Kabashima and Shinomoto 1992) that estimators of a binary decision boundary show asymptotically strange behaviors when the probability model is ill-posed or semiparametric. We give a rigorous analysis of this phenomenon in a stochastic perceptron by using the estimating function method. A stochastic perceptron consists of a neuron that is excited depending on the weighted sum of inputs but its probability distribution form is unknown here. It is shown that there exists no √n-consistent estimator of the threshold value h, that is, no estimator h that converges to h in the order of 1/ √n as the number n of observations increases. Therefore, the accuracy of estimation is much worse in this semiparametric case with an unspecified probability function than in the ordinary case. On the other hand, it is shown that there is a √n-consistent estimator ■ of the synaptic weight vector. These results elucidate strange behaviors of learning curves in a semiparametric statistical model.

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76. Degree of Approximation Results for Feedforward Networks Approximating Unknown Mappings and Their Derivatives

Kurt Hornik, Maxwell B. Stinchcombe, Halbert White, Peter Auer

https://doi.org/10.1162/neco.1994.6.6.1262 | pp. 1262-1275

Recently Barron (1993) has given rates for hidden layer feedforward networks with sigmoid activation functions approximating a class of functions satisfying a certain smoothness condition. These rates do not depend on the dimension of the input space. We extend Barron's results to feedforward networks with possibly nonsigmoid activation functions approximating mappings and their derivatives simultaneously. Our conditions are similar but not identical to Barron's, but we obtain the same rates of approximation, showing

that the approximation error decreases at rates as fast as n –1/2 , where n is the number of hidden units. The dimension of the input space appears only in the constants of our bounds.

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77. Influence Function Analysis of PCA and BCM Learning

Yong Liu

https://doi.org/10.1162/neco.1994.6.6.1276 | pp. 1276-1288

Based on the influence function (Hampel et al. 1986), we analyze several forms of learning rules under the influence of abnormal inputs (outliers). Principal component analysis (PCA) learning rules (Oja 1982, 1989, 1992) are shown to be sensitive to the influence of outliers. Biologically motivated robust PCA learning rules are proposed. The variants of BCM learning (Bienenstock et al. 1982; Intrator 1990b) with linear neurons are shown to be subject to the influence of outliers, in contrast they are shown to be robust to outliers with sigmoidal neurons.

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78. Boosting and Other Ensemble Methods

Harris Drucker, Corinna Cortes, L. D. Jackel, Yann LeCun, Vladimir Vapnik

https://doi.org/10.1162/neco.1994.6.6.1289 | pp. 1289-1301

We compare the performance of three types of neural network-based ensemble techniques to that of a single neural network. The ensemble algorithms are two versions of boosting and committees of neural networks trained independently. For each of the four algorithms, we experimentally determine the test and training error curves in an optical character recognition (OCR) problem as both a function of training set size and computational cost using three architectures. We show that a single machine is best for small training set size while for large training set size some version of boosting is best. However, for a given computational cost, boosting is always best. Furthermore, we show a surprising result for the original boosting algorithm: namely, that as the training set size increases, the training error decreases until it asymptotes to the test error rate. This has potential implications in the search for better training algorithms.

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79. Index to Volume 6

(authors unknown)

https://doi.org/10.1162/neco.1994.6.6.1303 | pp. 1303-1312

No abstract available.